

Cross-Layer Trajectory Instability for Single-Pass Hallucination Detection in Large Language Models

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Abstract

Large language models produce confident but factually incorrect outputs, yet existing detection methods require multiple forward passes or external knowledge bases. This work proposes Cross-Layer Trajectory Instability (CLTI), a single-pass approach that measures how token representations evolve across late transformer layers. On TruthfulQA MC1 with LLaMA-2-7B (layers 24–31), Normalized Trajectory Instability (NTI) achieves AUROC 0.5657 (5-fold CV, all folds > 0.52). Combining NTI with Convergence Monotonicity Index (CMI) and output entropy yields +7.1% AUROC improvement over an intercept-only baseline (Fisher combined $p = 1.15e-05$). However, trajectory metrics fail on low-entropy predictions (AUROC 0.5136, 95% CI [0.4639, 0.5628] includes chance), and Representational Competition Index (RCI) flip patterns occur in >90% of both correct and incorrect responses, indicating an architectural rather than epistemic phenomenon. These results demonstrate that trajectory features provide complementary signal for uncertain predictions while identifying constraints that limit the method’s scope.

1. Introduction

Large language models generate confident but factually incorrect statements, a phenomenon termed hallucination that undermines deployment in high-stakes domains. Detecting hallucinations before they reach users requires uncertainty quantification, yet current methods impose computational overhead. Semantic

entropy clusters multiple sampled outputs for linguistic invariance but requires 5-10 forward passes per query. Probe-based methods train classifiers on hidden states, adding inference latency. Single-pass approaches using final-layer output entropy ignore internal processing dynamics that may distinguish factual retrieval from fabrication.

This work hypothesizes that the trajectory of representations across late layers carries discriminative signal. Under the Cross-Layer Trajectory Instability (CLTI) framework, factual retrieval follows stable attractor dynamics—representations converge monotonically toward a knowledge-grounded answer. Fabrication, lacking grounded knowledge, requires iterative cross-layer constraint satisfaction, producing higher trajectory instability.

Three single-pass metrics computed from layers 24–31 (8 late layers) of LLaMA-2-7B on TruthfulQA MC1 operationalize this hypothesis:

- **Normalized Trajectory Instability (NTI):** variance of per-layer entropy normalized by mean entropy
- **Convergence Monotonicity Index (CMI):** proportion of layer transitions where answer similarity increases
- **Representational Competition Index (RCI):** top-token flip patterns across layers

Experiments validate the core claim while refuting secondary mechanisms. NTI achieves AUROC 0.5657 on 5-fold cross-validation, exceeding the 0.55 threshold specified a priori. Combined with CMI and final-layer entropy, detection improves by +7.1% AUROC (Fisher combined $p = 1.15e-05$) over an intercept-only baseline. However, trajectory metrics fail on low-entropy predictions (AUROC 0.5136, 95% CI includes chance), and RCI flip patterns appear in >90% of both correct and incorrect responses.

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1. Existence evidence that single-pass trajectory features discriminate hallucinations beyond output entropy
2. Mechanism refinement showing the signal is driven by high-uncertainty cases; confident hallucinations remain undetected
3. Negative results demonstrating RCI flip patterns are universal in transformer processing

2. Related Work

2.0.1. 2.1 UNCERTAINTY QUANTIFICATION IN LLMs

Early work on LLM uncertainty relies on token-level conditional probabilities, but RLHF-trained models exhibit poor calibration between confidence and accuracy. Verbalized confidence often outperforms raw probabilities for post-RLHF models.

Semantic entropy addresses linguistic invariance by clustering semantically equivalent generations before computing entropy, achieving AUROC approximately 0.75 on TruthfulQA but requiring 5-10 samples per query. SelfCheckGPT provides black-box consistency checking via sampling and cross-referencing, again requiring multiple generations.

Probe-based methods avoid multi-sampling by training classifiers on internal representations. MIND uses internal states for real-time unsupervised detection. Semantic entropy probes demonstrate that lightweight linear classifiers on hidden states approach the accuracy of full semantic entropy computation.

2.0.2. 2.2 CROSS-LAYER ANALYSIS

The logit lens projects intermediate representations through the final unembedding matrix, revealing per-layer probability distributions. END decoding demonstrates that cross-layer entropy correlates with factuality. The present work formalizes this observation into trajectory metrics (NTI, CMI) and validates them with controlled experiments.

2.0.3. 2.3 POSITIONING

Method	Single-Pass	No External DB	Interpretable
Semantic Entropy	No	Yes	Moderate
SelfCheckGPT	No	Yes	High
MIND	Yes	Yes	Low
CLTI (this work)	Yes	Yes	High

3. Methodology

3.0.1. 3.1 TRAJECTORY METRICS

Normalized Trajectory Instability (NTI):

$$NTI = \frac{\sigma(\{H_l\}_{l=24}^{31})}{\mu(\{H_l\}_{l=24}^{31}) + \epsilon}$$

where H_l denotes the entropy of the logit distribution at layer l .

Convergence Monotonicity Index (CMI):

$$CMI = \frac{1}{L-1} \sum_{l=24}^{30} \mathbb{I}[s_{l+1} > s_l - \delta]$$

where s_l denotes similarity between the layer- l representation and the correct answer token.

Representational Competition Index (RCI): Binary indicator of top-token change between consecutive layers.

3.0.2. 3.2 MODEL AND DATASET

- **Model:** LLaMA-2-7B (meta-llama/Llama-2-7b-hf)
- **Dataset:** TruthfulQA MC1 (817 questions, 4114 prompt-choice pairs)
- **Evaluation:** 5-fold stratified cross-validation, logistic regression ($C=1.0$), AUROC
- **Random seed:** 42

3.0.3. 3.3 SUB-HYPOTHESES

ID	Hypothesis	Success Criterion	Gate
h-e1	NTI AUROC > 0.55	Mean > 0.55, all folds > 0.52	MUST_WORK
h-m1	Combined gain ≥ 0.03	LRT p < 0.05	SHOULD_WORK
h-m2	Low-entropy AUROC > 0.55	CI LB > 0.50	SHOULD_WORK
h-m3	RCI flip separation $\geq 20\%$	$\geq 30\%$ halluc, <10% correct	SHOULD_WORK

4. Experimental Setup

Feature extraction used layers 24–31 of LLaMA-2-7B on 4114 prompt-choice pairs from TruthfulQA MC1. For each pair, the model processed the prompt concatenated with each answer choice, extracting hidden states at each layer. Per-layer entropy was computed by projecting hidden states through the unembedding matrix and applying softmax.

For h-m2 (low-entropy subset), the 25th percentile threshold was computed as $H_L < 0.0091$, yielding 1029 samples. Bootstrap confidence intervals used 1000 iterations with seed 42.

5. Results

5.0.1. 5.1 H-E1: NTI EXISTENCE (VALIDATED)

Metric	Value	Status
Mean AUROC	0.5657	PASS (> 0.55)
Fold 1	0.5356	> 0.52
Fold 2	0.5954	> 0.52
Fold 3	0.5665	> 0.52
Fold 4	0.5469	> 0.52
Fold 5	0.5839	> 0.52

5.0.2. 5.2 H-M1: COMBINED MODEL (VALIDATED)

Metric	Value
Intercept-only baseline	0.5000
Full model (H_L +NTI+CMI)	0.5712
AUROC Gain	+0.0712
Fisher combined p-value	1.15e-05

Per-fold gains: 0.0645, 0.1062, 0.0747, 0.0381, 0.0727 (mean = 0.0712).

When entropy (H_L) is the sole predictor, logistic regression coefficients fail to converge, producing intercept-only predictions (AUROC = 0.50 by construction). The reported gain reflects trajectory features’ incremental discriminative validity.

5.0.3. 5.3 H-M2: LOW-ENTROPY SUBSET (REFUTED)

Metric	Value
Subset size	1029 (25th percentile)
H_L threshold	0.0091
AUROC	0.5136
95% CI	[0.4639, 0.5628]

The 95% confidence interval includes 0.50, indicating no reliable signal on low-entropy (“confident”) predictions.

5.0.4. 5.4 H-M3: RCI FLIP PATTERN (REFUTED)

The flip pattern is near-universal across both classes, indicating an architectural phenomenon rather than an epistemic signal.

Class	Flip Rate
Hallucinations	95.1%
Correct	90.9%
Separation	4.2%

6. Discussion

6.0.1. VALIDATED CLAIMS

NTI combined with CMI improves detection by +7.1% AUROC over an entropy-only baseline (Fisher combined $p = 1.15e-05$). Trajectory instability correlates with hallucination on high-entropy samples where the model exhibits uncertainty.

6.0.2. REFUTED CLAIMS

h-m2: The signal collapses on confident predictions. NTI is coupled to entropy by construction; when final-layer entropy is low, trajectory variance is also constrained.

h-m3: RCI flip is architectural, not epistemic. Top-token changes between layers occur in $>90\%$ of all responses regardless of correctness. This pattern reflects normal transformer processing rather than uncertainty-specific dynamics.

6.0.3. LIMITATIONS

- Entropy coupling:** The method cannot detect confident hallucinations—cases where the model is wrong but not uncertain.
- Single model:** Results are from LLaMA-2-7B only; generalization to other architectures and scales requires replication.
- Multiple-choice format:** TruthfulQA MC1 uses forced-choice evaluation; behavior on free-form generation may differ.
- Correlational evidence:** No causal intervention was performed; the observed patterns may not reflect the mechanism of hallucination.

6.0.4. FUTURE WORK

- Develop entropy-orthogonal features using hidden state geometry
- Validate on additional models (Mistral, LLaMA-3)

- Investigate semantic RCI measuring distance between competing tokens rather than binary flips
- Apply causal interventions through activation patching

7. Conclusion

This work introduced Cross-Layer Trajectory Instability (CLTI) for single-pass hallucination detection. NTI achieves AUROC 0.5657 on TruthfulQA MC1; combined with CMI and entropy, +7.1% improvement is observed (Fisher combined $p = 1.15e-05$). Trajectory metrics fail on low-entropy predictions (AUROC 0.5136, CI includes chance), and RCI flip patterns are architectural. The method complements entropy-based detection on uncertain cases; confident hallucinations remain an open problem.

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